DATA STRUCTURES

LAB 14

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GULBERG GREEN ISLAMABAD

Lab Task 1

Assume Graph

Vertices: A, B, C, D
Edges: A–B, A–C, B–D, C–D

1. Adjacency Matrix

	A	B	C	D
A	0	1	1	0
B	1	0	0	1
C	1	0	0	1
D	0	1	1	0

2. Adjacency List

- A → B, C
- B → A, D
- C → A, D
- D → B, C

3. Breadth First Search (BFS)

Starting from vertex A:

Traversal order:
A → B → C → D

4. Depth First Search (DFS)

Starting from vertex A:

Traversal order:
A → B → D → C

<u>Lab Task 2</u>

We model airline routes using graph theory.

Vertices (Cities)

Boston, Newark, Miami, Detroit, Washington

Case 1: Simple Graph

- One edge between cities if at least one flight exists (ignore direction and number of flights).
- Type: **Undirected Simple Graph**

Edges:

- Boston — Newark
- Newark — Miami
- Newark — Detroit
- Newark — Washington
- Washington — Miami

Case 2: Multigraph

- Multiple edges represent multiple flights.
- Type: **Undirected Multigraph**

Edges:

- Boston ↔ Newark (6 edges total: 4 + 2)
- Newark ↔ Miami (5 edges: 3 + 2)
- Newark ↔ Detroit (3 edges: 1 + 2)
- Newark ↔ Washington (5 edges: 3 + 2)
- Washington ↔ Miami (1 edge)

Case 3: Multigraph with Loop

- Same as Case 2 plus a loop.
- Add:
    - One loop at Miami (sightseeing trip)

Type: **Undirected Multigraph with Loop (Pseudograph)**

Lab Task 3

Problem

Model relationships where each person knows the name of another.

Graph Model

Vertices

Each vertex represents a student in the class.

Edges

An edge represents “knows the name of”.

Analysis

- **Directed or Undirected?**
  Directed, because knowing someone’s name may not be mutual.
- **Multiple Edges?**
  Not required. One edge is enough to represent the relationship.
- **Loops?**
  Not required, because a person knowing their own name is trivial and usually ignored.

Final Model

- Type: **Directed Simple Graph**

Lab Task 4

Using the same sample graph from Task 1:

Vertices: A, B, C, D

BFS Traversal

Starting from A:
A → B → C → D

DFS Traversal

Starting from A:
A → B → D → C

THANK YOU!!!!